

Unit 2: Characteristics and Classification

Complete Study Notes – Grade 9 Biology

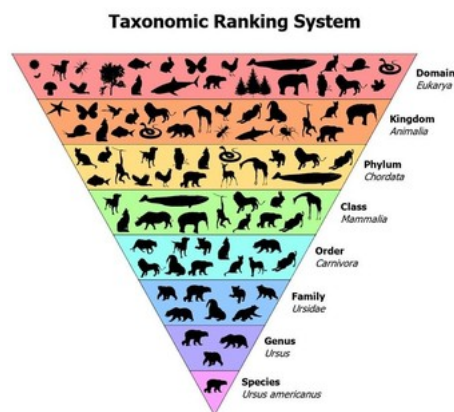


Figure 6: Taxonomic hierarchy – domain, kingdom, phylum, class, order, family, genus, species.

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2.1 Characteristics of Living Things

Living things have a variety of shapes and forms. Biologists study life in many different ways. What makes something 'alive'? Anyone could deduce that a galloping horse is alive and a car is not, but why? We cannot say 'If it moves, it's alive' because a car can move, and gelatin can wiggle in a bowl. Although we cannot define life with a single sentence, we can identify a series of characteristics shared by all living systems.

Properties shared by all living things:

- **All living things are made up of one or more cells** – Those made up of one cell, such as bacteria, are termed *unicellular*; those made up of more than one cell, such as plants and animals, are termed *multicellular*.
- **All living things require energy** – All organisms use a source of energy for their metabolic activities. Plants are *autotrophs* (producers) that make their own food through photosynthesis. Animals are *heterotrophs* (consumers) that obtain energy by consuming other organisms.
- **All living organisms respond to stimuli** – Organisms can detect or sense stimuli (changes) in the internal or external environment and make appropriate responses. Example: plants grow toward light (phototropism).
- **All living things can grow** – Growth is a permanent increase in size and mass due to an increase in cell number or cell size (or both). Even bacteria show an increase in size. Multicellular organisms increase the number of cells in their bodies, becoming more complex and changing their shape and size.
- **All living things can reproduce** – Reproduction is the process that makes more of the same kind of organism. Single-celled organisms may simply divide into two (binary fission). Multicellular plants and animals may reproduce sexually or asexually.
- **All living things can excrete** – Excretion is the removal of metabolic wastes produced in cells as a result of chemical reactions (metabolism). Example: carbon dioxide produced by respiration is expelled from the body.
- **All living things display ordered complexity** – All living things are both complex and highly ordered. The levels of organization begin with atoms and molecules and increase in complexity up to the organism level.
- **Most living things maintain homeostasis** – Most organisms maintain relatively constant internal conditions that are different from their environment. Homeostasis is the regulation of an organism's internal conditions to maintain stability. Example: human body temperature remains stable despite changes in outside temperatures.
- **All living things possess adaptations that evolve over time** – All organisms interact with other organisms and their environment in ways that influence their survival, and as a result, organisms evolve adaptations to their environments.

Example – Distinguishing Living from Non-living: A car moves, but it does not grow, reproduce, or respond to stimuli in a biological sense. A crystal grows, but it does not reproduce or maintain homeostasis. Living things possess *all* of the above characteristics, not just one or two.

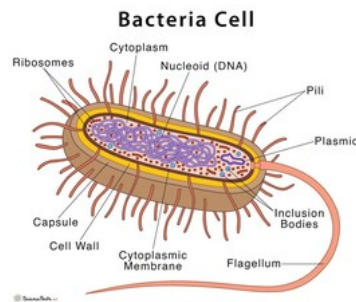


Figure 1: Bacteria (Kingdom Monera) – prokaryotic, unicellular organisms.

2.2 Taxonomy of Living Things

2.2.1 Principles of Classification

Organizing items not only makes them easier to find but can also make them easier to understand. One tool biologists use to organize and understand living organisms is **classification** – the process of grouping things based on their similarities.

The science of naming, identifying, and classifying organisms is known as **taxonomy**. Scientists who study taxonomy are called **taxonomists**.

Biologists classify organisms into different categories mostly by judging the degrees of their apparent similarities and differences. These include:

- **Morphological characters** – external and internal structures of the organism.
- **Physiological characters** – how the organism functions (metabolism, reproduction).
- **Molecular characters** – genetic makeup (DNA, RNA) to reveal evolutionary relationships.
- **Behavioural characters** – how the organism behaves.
- **Ecological characters** – where and how the organism lives in its environment.

The assumption is that the greater the degree of physical similarity between organisms, the closer their biological relationship is.

2.2.2 Taxonomic Hierarchies in Biological Classification

Thousands of years ago, the Greek philosopher **Aristotle** (384–322 BC) developed the first widely accepted biological classification systems. He used simple morphological characters to classify plants into trees, shrubs, and herbs. He also divided animals into two groups – those which had red blood and those that did not.

Later, in the 1700s, a Swedish botanist **Carolus Linnaeus** (1707–1778), also known as the *father of taxonomy*, introduced a taxonomic hierarchy of classification. He was the first person to propose an orderly system for classifying organisms.

Taxonomic Hierarchy is the process of arranging various organisms into successive levels of biological classification either in decreasing or increasing order.

In the Linnaean classification system, all organisms are placed in a ranked hierarchy (from broadest to most specific):

- **Domain** – The broadest level. Three domains: Bacteria, Archaea, Eukarya.
- **Kingdom** – A taxon of similar phyla/divisions. Example: Animalia, Plantae.

- **Phylum (Division in plants)** – A taxon of similar classes. Example: Chordata.
- **Class** – A taxon of similar orders. Example: Mammalia.
- **Order** – A taxon of similar families. Example: Primates.
- **Family** – A taxon of similar genera. Example: Hominidae.
- **Genus (plural: genera)** – A taxon of similar species. Example: *Homo*.
- **Species** – The most specific level. A unique type of organism. Example: *Homo sapiens*.

The Three Domains:

- **Bacteria** – Prokaryotic, unicellular, have cell walls with peptidoglycan, live in most habitats.
- **Archaea** – Prokaryotic, unicellular, have very different cell walls (no peptidoglycan), live in extreme environments (hot springs, salt lakes).
- **Eukarya** – Eukaryotic cells (have a nucleus and membrane-bound organelles). Includes protists, fungi, plants, and animals.

Example – Human Classification: Domain: Eukarya, Kingdom: Animalia, Phylum: Chordata, Class: Mammalia, Order: Primates, Family: Hominidae, Genus: *Homo*, Species: *Homo sapiens*.

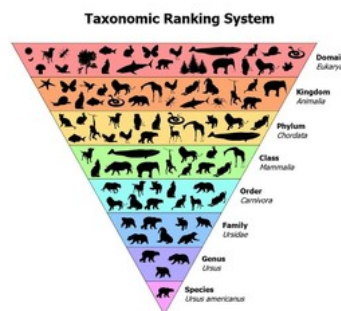


Figure 6: Taxonomic hierarchy – domain, kingdom, phylum, class, order, family, genus, species.

2.3 Relevance of Classification

The applications of classification are manifold:

- **Identification of harmful and beneficial organisms** – Helps distinguish pathogens from useful organisms.
- **Reduce energy and time to study organisms** – By organising organisms into groups, studying one representative can provide insights about the whole group.
- **Help understand the relationship between organisms** – Classification gives biologists a framework to study evolutionary relationships. For example, it has helped scientists understand that birds are more closely related to dinosaurs than to reptiles.
- **Useful tool for agriculture, forestry, and medicine** – Taxonomists can discover new sources of lumber, foods, medicines, and energy. Example: If one species of tree contains useful chemicals (like Moringa), a closely related species might have similar uses.
- **Helps the economy** – Discovery of new species with economic value (food, medicine, materials).

Example – Ethiopian Context: *Moringa stenopetala* (Shiferaw) is known for its medicinal and nutritional value. Classification helps identify other plant species that might have similar properties.



Figure 7: Common Ethiopian plants – tef, noug, enset, and coffee.

2.4 Linnaean System of Nomenclature

Linnaeus refined and popularized the **binomial system of naming organisms**, in which the first name represents the *genus* and the second name represents the *specific epithet*. This system is still the official starting point for naming or revising the names of organisms.

Why Binomial Nomenclature?

Species must be named in such a way that the name is recognised all over the world. For example, maize is named *Bekolo* in Amharic, *Boqqollo* in Afan Oromo, *Baddela* in Sidaamu Afoo, and *corn* in American English. If you are not aware of these names, it could lead to confusion. The botanical name, *Zea mays*, is recognised worldwide.

The Latin form of the name allows it to be used all over the world irrespective of language barriers. The similarity in size, shape, behaviour, and colour makes it difficult to differentiate one species from another. The use of the scientific name avoids such confusion.

Rules for Writing Scientific Names:

- The first letter of the genus is **capitalised**.
- The specific epithet always starts with a **small letter**.
- The scientific name is **underlined** when handwritten or **italicised** when printed.
- Example: *Homo sapiens* (human beings).

Examples of Scientific Names:

- House fly – *Musca domestica*
- Dog – *Canis familiaris*
- Cat – *Felis catus*
- Goat – *Capra hircus*
- Mouse – *Mus musculus*
- Maize – *Zea mays*
- Tef – *Eragrostis tef*
- Noug – *Guizotia abyssinica*
- Enset (false banana) – *Ensete ventricosum*

2.5 Common Ethiopian Animals and Plants

Ethiopia is endowed with high biological diversity (biodiversity) due to its geographical location, topographical diversity, and diverse climatic features. The country is a hotspot for a diversity of wild plant and animal species with a high degree of **endemism** (species found nowhere else).

Furthermore, Ethiopia is a primary centre of diversity for field crops such as:

- Noug (*Guizotia abyssinica*) – Ethiopia is the primary centre of origin.
- Tef (*Eragrostis tef*) – The only cultivated species of the genus.
- Ethiopian mustard (*Brassica carinata*) – Used for oil and food.
- Barley, sorghum, durum wheat, finger millet, faba bean, chickpea, lentil, and cowpea – all have wide diversity in Ethiopia.

Ethiopia has also served as a gateway for domestic animals from Asia to Africa, and its diverse ecology has favoured the diversification of these resources.

Dichotomous Keys

Dichotomous keys are used to identify unfamiliar organisms. They simplify the process of identification. Each key is made up of pairs of contrasting features (dichotomous means *two branches*), starting with quite general characteristics and progressing to more specific ones.

How to use a dichotomous key:

- Item 1 gives a choice between two alternatives (e.g., warm-blooded vs cold-blooded).
- If the animal is cold-blooded, move to Item 2 and make a further choice.
- If it is warm-blooded, move to Item 4 for your next choice.
- Continue making choices until the organism is identified.

Example: A key might ask: 'Does the animal have feathers?' If yes → bird. If no → 'Does it have fur?' If yes → mammal.

Common Ethiopian Animals and Their Scientific Names:

- Elephant – *Loxodonta africana* (Mammal)
- Ethiopian Wolf – *Canis simensis* (Mammal – endemic to Ethiopia)
- Gelada – *Theropithecus gelada* (Mammal – endemic to Ethiopia)
- Lion – *Panthera leo* (Mammal)
- Walia Ibex – *Capra walie* (Mammal – endemic to Ethiopia)
- Ostrich – *Struthio camelus* (Bird)
- Wattled Ibis – *Bostrychia carunculata* (Bird – endemic to Ethiopia)

Common Ethiopian Plants and Their Scientific Names:

- Enset – *Ensete ventricosum* (False banana – staple food in southern Ethiopia)
- Maize – *Zea mays* (Major cereal crop)
- Noug – *Guizotia abyssinica* (Oilseed crop – endemic to Ethiopia)
- Tef – *Eragrostis tef* (Major cereal – endemic to Ethiopia)
- Wheat – *Triticum aestivum* (Major cereal crop)
- Coffee – *Coffea arabica* (Native to Ethiopia – important cash crop)



Figure 7: Common Ethiopian plants – tef, noug, enset, and coffee.

2.6 The Five-Kingdom System of Classification

Classification of organisms into plants and animals was easily done and understood, but a large number of organisms did not fall into either category. Hence, the two-kingdom classification used for a long time was found inadequate.

Besides gross morphology, a need was also felt for including other characteristics like cell structure, nature of cell wall, mode of nutrition, habitat, methods of reproduction, and evolutionary relationships.

Whittaker (1969) proposed a **five-kingdom classification** to solve the pitfalls of the two-kingdom system. The main criteria for classification used by him include:

- Cell structure (prokaryotic vs eukaryotic)
- Body organisation (unicellular vs multicellular)
- Mode of nutrition (autotrophic vs heterotrophic)
- Reproduction and phylogenetic relationships

Whittaker's five kingdoms are: **Monera**, **Protista**, **Fungi**, **Plantae**, and **Animalia**.

Note about Viruses: Viruses are not included in any kingdom – they are not considered living organisms because they lack key characteristics of living things. Viruses are particles that are not alive. They are made up of nucleic acids (DNA or RNA) surrounded by a protein coat. They need a host cell to replicate.

2.6.1 Kingdom Monera

Monera includes **Eubacteria** (true bacteria) and **Archaeobacteria** (Archaea).

Characteristics of Monera:

- Simple **prokaryotic** unicellular organisms.
- Lack a well-defined nucleus or any membrane-bound organelles.
- Most have a **rigid cell wall**.
- Have various modes of nutrition – could be **autotrophs** (photosynthetic or chemosynthetic) or **heterotrophs**.
- Known as **decomposers** and **mineralizers** in the biosphere.

Eubacteria (True Bacteria):

- Have strong cell walls. Exist in various shapes and forms (cocci, bacilli, spirilla).

- Some are heterotrophs; others can make their food (autotrophs). Some autotrophic bacteria are photosynthetic (like cyanobacteria); others are chemosynthetic.
- Live in most habitats, except the most extreme.
- Some cause diseases (e.g., *Mycobacterium tuberculosis* causes tuberculosis, *Haemophilus ducreyi* causes chancroid). Most, however, are harmless and helpful.

Archaeobacteria (Archaea):

- Have very different cell walls from bacteria.
- Can be chemosynthetic or photosynthetic.
- Live in extreme environments – swamps, deep-ocean hot-water vents, seawater evaporating ponds, often with no oxygen.

Importance of Bacteria:

- **Beneficial:** Help with human digestion, produce cheese, yogurt, sourdough bread, fix nitrogen in soil, decompose organic matter, produce oxygen (cyanobacteria).
- **Harmful:** Cause diseases (tuberculosis, pneumonia, strep throat, cholera), produce toxins that damage tissues.

Example: *E. coli* is a bacterium found in the human intestine. Most strains are harmless and help with digestion, but some strains can cause food poisoning.

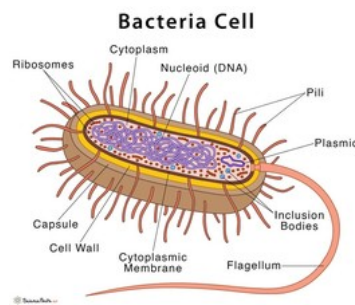


Figure 1: Bacteria (Kingdom Monera) – prokaryotic, unicellular organisms.

2.6.2 Kingdom Protista

Kingdom Protista contains the most diverse organisms of all the kingdoms.

Characteristics of Protists:

- All are **eukaryotic** – have a nucleus and membrane-bound organelles.
- Most are **unicellular**, but there are multicellular and colonial species.
- Some are **autotrophs** (like algae – photosynthetic).
- Others are **heterotrophs** (like protozoans – ingest food).
- Some are **animal-like** (protozoans), some are **plant-like** (algae), and some are **fungus-like** (slime moulds).

Types of Protists:

- **Protozoans** – Animal-like protists. Examples: *Amoeba*, *Paramecium*, *Plasmodium* (causes malaria).
- **Algae** – Plant-like protists. Examples: *Spirogyra*, *Chlamydomonas*, diatoms. They are photosynthetic and form the basis of aquatic food chains. Produce much of the oxygen in Earth's atmosphere.

- **Fungus-like protists** – Examples: slime moulds, water moulds.

Importance of Protists:

- **Ecological:** Algae are primary producers in aquatic ecosystems; produce oxygen.
- **Economic:** Diatomite (from diatom remains) is used in pet litter, paint, and abrasives.
- **Medical:** *Plasmodium* causes malaria; *Trypanosoma* causes sleeping sickness.

Example: Diatoms are a type of algae whose remains become a powdery, porous rock called diatomite. Diatomite is highly absorbent and is used in pet litter, chemical spill cleanup, and as an abrasive in household cleaners.

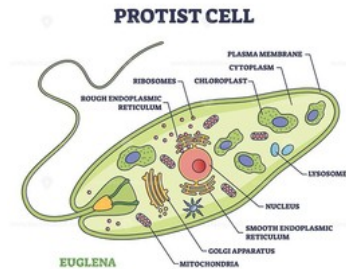


Figure 2: Protists (Kingdom Protista) – diverse eukaryotic organisms.

2.6.3 Kingdom Fungi

Fungi are eukaryotic organisms that include microorganisms such as yeasts, moulds, and mushrooms.

Characteristics of Fungi:

- Eukaryotic, mostly **multicellular** (except yeasts, which are unicellular).
- Have cell walls made of **chitin** and polysaccharides.
- **Heterotrophic** – obtain nutrients by **absorption** (extracellular digestion).
- Secrete powerful enzymes into their surroundings to break down complex molecules into smaller organic compounds that they can absorb.
- Body consists of long, slender, thread-like structures called **hyphae** (a network of hyphae is called a **mycelium**).
- Can be **saprophytes** (decomposers) – absorb nutrients from dead organic matter.
- Can be **parasites** – absorb nutrients from living hosts (cause diseases like athlete's foot, ringworm, Candida infections).
- Can be **mutualists** – form symbiotic relationships (e.g., lichens with algae, mycorrhizae with plant roots).

Reproduction in Fungi:

- **Asexual reproduction:** Vegetative means (fragmentation, fission, budding) or spore formation.
- **Sexual reproduction:** Produces spores in distinct structures called **fruiting bodies**.

Importance of Fungi:

- **Ecological:** Decomposers – break down dead organic matter, recycling nutrients. Without fungi, the world would be overrun with waste.

- **Economic:** Yeast (*Saccharomyces cerevisiae*) is used in baking (making bread rise) and alcohol production (Tej, Tella, beer, wine).
- **Medical:** Penicillin is produced by the mould fungus *Penicillium*. Some fungi cause diseases.
- **Environmental:** Mycorrhizal fungi help plants absorb nutrients; lichens are pioneer species in primary succession.

Example: Yeast (*Saccharomyces cerevisiae*) is a unicellular fungus used to make *injera* rise and to produce alcoholic beverages like Tej and Tella. It carries out fermentation, converting sugars into alcohol and carbon dioxide.

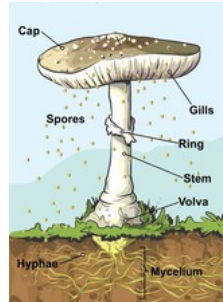


Figure 3: Fungi – eukaryotic, heterotrophic organisms with chitin cell walls.

2.6.4 Kingdom Plantae

Kingdom Plantae includes all eukaryotic, multicellular, autotrophic organisms.

Characteristics of Plants:

- **Eukaryotic** and **multicellular**.
- **Autotrophic** – make their own food through **photosynthesis** using chlorophyll.
- Have **cell walls** made of **cellulose**.
- **Stationary** – do not move from place to place.
- Reproduce both asexually (vegetative propagation) and sexually.
- Provide oxygen through photosynthesis.

Major Groups of Plants:

- **Bryophytes** – Non-vascular plants (mosses, liverworts). Have no vascular tissues, seedless.
- **Pteridophytes** – Vascular plants, seedless (ferns).
- **Gymnosperms** – Vascular plants, 'naked' seeds (conifers, pines, cycads).
- **Angiosperms** – Vascular plants, seeds enclosed in fruits (flowering plants). Angiosperms are divided into **monocots** (one seed leaf) and **dicots** (two seed leaves).

Economic Importance of Plants:

- **Food:** Provide staple crops (tef, wheat, maize, enset, fruits, vegetables).
- **Medicine:** Many plants have medicinal properties (e.g., Moringa, Artemisia).
- **Oxygen:** Produce oxygen through photosynthesis.
- **Habitat:** Provide shelter and food for animals.
- **Timber and fuel:** Provide wood for construction and firewood.
- **Cultural/spiritual:** Used in ceremonies, decorations, and religious practices.

Example: *Ensete ventricosum* (enset or false banana) is a staple food for millions of people in southern Ethiopia. It is drought-resistant and has high cultural and economic value.



Figure 4: Plants (Kingdom Plantae) – autotrophic, multicellular eukaryotes.

2.6.5 Kingdom Animalia

Kingdom Animalia includes all multicellular, heterotrophic, eukaryotic organisms.

Characteristics of Animals:

- **Eukaryotic** and **multicellular**.
- **Heterotrophic** – obtain food by **ingestion** (eat other organisms) and then digest it inside the body.
- **Lack cell walls** – animal cells are held together by structural proteins (most abundant is collagen).
- Have specialized cells: **muscle cells** (for movement) and **nerve cells** (for conducting impulses).
- Most reproduce **sexually**.
- Very diverse – classified into **invertebrates** (no backbone) and **vertebrates** (have a backbone).

Major Groups of Invertebrates:

- **Sponges** – simplest animals, sessile, filter feeders.
- **Cnidarians** – jellyfish, corals, sea anemones (have stinging cells).
- **Flatworms** – tapeworms, planarians (flattened bodies).
- **Roundworms** – nematodes (unsegmented, round bodies).
- **Annelids** – segmented worms (earthworms, leeches).
- **Molluscs** – snails, clams, octopuses (soft bodies, often with shells).
- **Arthropods** – insects, arachnids, crustaceans, myriapods (jointed legs, exoskeleton).
- **Echinoderms** – starfish, sea urchins (spiny skin).

Major Groups of Vertebrates:

- **Fish** – Aquatic, gills, scales (e.g., Nile tilapia, catfish).
- **Amphibians** – Live in water and on land, moist skin (e.g., frogs, toads).
- **Reptiles** – Dry scaly skin, lay eggs on land (e.g., lizards, snakes, crocodiles).
- **Birds** – Feathers, wings, lay eggs (e.g., ostrich, wattled ibis).
- **Mammals** – Hair, produce milk, give birth to live young (e.g., lion, elephant, Ethiopian wolf).

Economic and Ecological Importance of Animals:

- **Food:** Provide meat, milk, eggs, fish.
- **Pollination:** Insects (bees) pollinate crops.

- **Pest control:** Many animals eat pests.
- **Scientific research:** Used in medicine and biology.
- **Ecotourism:** Wildlife attracts tourists (e.g., Ethiopian wolves, geladas).
- **Cultural/spiritual:** Used in traditions and ceremonies.

Example: The Ethiopian wolf (*Canis simensis*) is endemic to Ethiopia and is found only in the highlands. It is the most endangered carnivore in Africa and is an important species for ecotourism and conservation.



Figure 5: Animals (Kingdom Animalia) – heterotrophic, multicellular eukaryotes.

2.7 Renowned Taxonomists in Ethiopia

Ethiopia is known for high species diversity and endemism. However, only a few studies have been done to identify, name, and classify the biodiversity at different levels. Among the factors that led to inadequate levels of studies could be the few number of scientists educated and trained in taxonomy despite the country's rich biodiversity.

Yet, as the result of efforts made over the past six decades, our country has trained and educated a couple of renowned taxonomists that contributed to the publication of volumes of books on the Flora of Ethiopia, for example.

Notable Ethiopian Taxonomists:

- **Dr. Mesfin Taddese** – Contributed to the study of Ethiopian flora and vegetation.
- **Professor Sebsebe Demissew** – Has participated in several successful research projects with universities in Europe and Africa. Published books and articles on the vegetation and plants of Ethiopia and Africa. Served as Chair of the Biological Society of Ethiopia, Secretary-General of AETFAT, Director of Flora of Ethiopia and Gullele Botanical Garden.
- **Professor Ensermu Kelbessa** – Contributed significantly to plant taxonomy in Ethiopia.
- **Professor Silesh Nemomissa** – A well-known plant taxonomist.
- **Professor Abebe Getahun** – Contributed to the field of animal taxonomy, particularly fish taxonomy in Ethiopia.

Example – Professor Sebsebe Demissew: He is an internationally recognised botanist. He has published extensively on Ethiopian plants and has been instrumental in establishing and developing the Gullele Botanical Garden in Addis Ababa, which serves as a centre for plant conservation and research.

Unit 2 Summary

- The characteristics of living things are movement, respiration, sensitivity, growth, reproduction, excretion, nutrition, homeostasis, and evolutionary adaptation.

- Classification is a way of sorting organisms into a meaningful order, traditionally using morphology and anatomy, but recently also using DNA (molecular) evidence.
- The science of naming, identifying, and classifying organisms is known as **taxonomy**.
- Carolus Linnaeus (1707–1778), the 'father of taxonomy', introduced the taxonomic hierarchy: Domain, Kingdom, Phylum/Division, Class, Order, Family, Genus, Species.
- The three domains are Bacteria, Archaea, and Eukarya (protists, fungi, plants, and animals).
- The binomial system is an internationally agreed system in which the scientific name of an organism is made up of the genus and the specific epithet (e.g., *Homo sapiens*).
- Ethiopia is a primary centre of diversity for tef (*Eragrostis tef*), noug (*Guizotia abyssinica*), and Ethiopian mustard (*Brassica carinata*).
- Dichotomous keys are used to identify unfamiliar organisms using pairs of contrasting features.
- Whittaker's five-kingdom scheme consists of Monera, Protista, Fungi, Plantae, and Animalia.
- Monera (bacteria and archaea) are prokaryotic, unicellular, with no nucleus.
- Protists are eukaryotic, diverse (unicellular and multicellular), with various modes of nutrition.
- Fungi are eukaryotic heterotrophs that absorb nutrients after extracellular digestion. Have cell walls with chitin.
- Plantae are eukaryotic, multicellular, autotrophic organisms with cellulose cell walls.
- Animalia are eukaryotic, multicellular, heterotrophic organisms without cell walls.
- Renowned Ethiopian taxonomists include Professor Sebsebe Demissew, Professor Ensermu Kelbessa, Professor Silesh Nemomissa, Dr. Mesfin Taddese, and Professor Abebe Getahun.

End of Unit 2 – Characteristics and Classification of Organisms

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